

# OligoTox-Thrombocytopenia — Methodology

Materials and methods · NIH/NCATS OligoTox Open Data Challenge, Phase 2

**Nature of the dataset:** in-silico curation of pre-existing, publicly reported data. The "materials and methods" below therefore describe source identification, extraction, harmonisation, grading, provenance and quality control — how the dataset was assembled and computationally processed.

## 1. Materials — the sources

The material of this dataset is **70 primary sources**, each retrieved in full and cited to an exact locus. Four classes, in descending order of redistribution freedom:

| Class                           | Examples                                                                                                                                                                                                                                                                    | Retrieval route                                                                                                                  | rows                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>FDA documents</b>            | FDA prescribing information; FDA multidiscipline, clinical and pharmacology/toxicology reviews (NDA 211172, 217779, 203568, 210922, 212194, 214012, 215515, 206488, 217225); EMA SmPCs and EPAR assessment reports (Waylivra, Tegsedi, Onpattro, Oxlumo, Amvuttra, Qalsody) | DailyMed SPL REST API (structured XML); accessdata.fda.gov (browser User-Agent required); ema.europa.eu PDFs parsed with PyMuPDF | 285<br>FDA +<br>216<br>EMA |
| <b>USPTO patents</b>            | Ionis panels: US 2023/0263822, US 11,535,849, US 2021/0355497, US 2015/0031747, US 10,927,377, US 10,100,310                                                                                                                                                                | Google Patents HTML (USPTO print PDFs are image-only); examples tables and formal sequence listings                              | 241                        |
| <b>Peer-reviewed literature</b> | Sewing 2017 (PLOS); Valenzuela 2023 (MDPI); the ASO adverse-event meta-analysis; Crooke 2017 pooled safety database; Lundberg Slingsby 2022; Flierl 2015; Jaax 2013; Zaslavsky 2021; clinical trial reports                                                                 | Europe PMC / NCBI full-text XML plus supplementary files; PDF render where XML is abstract-only                                  | 1,211                      |
| <b>Trial registry</b>           | ClinicalTrials.gov posted results                                                                                                                                                                                                                                           | ClinicalTrials.gov API v2 (structured adverse-event and outcome-measure tables)                                                  | 6                          |

**These are acquisition classes, not rights classes.** An earlier version of this table labelled its rows "public domain" and "summary statistics", which asserted a rights position from the retrieval route. Where a document came from does not settle what may be reused from it: "regulator document = government work = public domain" reaches **US federal agencies only**, and EMA material is reusable with attribution rather than unrestricted. Rights are classified per publisher and per regulator in `curation/rights/shipped_row_rights.csv` and summarised in the PADP; 224 rows sit on a proposed hold awaiting the data owner's ruling.

Sources were discovered by a parallel sweep across ten independent lanes and ranked by **extractable per-measurement content**, not topicality. Each had to be actually fetched before being listed, so the registry contains no citation that cannot be re-checked. Regulatory review documents proved the richest and least-mined: they carry per-patient nadir listings, antibody cross-tabulations and dose-banded incidence grids absent from the corresponding journal papers.

## 2. Oligonucleotide identity, chemical modification and purity

### 2.1 How oligo identity was established and characterised

Because compounds were **not synthesised for this work**, "characterisation of identity" means establishing, from the primary record, exactly which molecule each row refers to. Four methods were used, each leaving an auditable trail in `design_source`:

- Patent sequence listings.** Formal listings give sequence and declared length. Every sequence was accepted only after matching an *independently declared* length — the LENGTH field, the target start/stop span, or the gapmer-motif sum. Across one patent, 2,631 sequences passed this dual check with zero conflicts.
- Deterministic parse of regulatory chemical names.** An FDA label's IUPAC chemical name spells out every residue. Parsing olezarsen's yielded `AGCTTCTTGTCAGCTTTAT`, and the stated molecular formula `C296H419N710154P20S19Na20` independently

fixes 19 phosphorothioate linkages across 20 residues — an orthogonal confirmation of both sequence length and backbone.

3. **WHO INN nomenclature.** INN lists spell out each residue longhand, permitting recovery by deterministic parse rather than transcription.
4. **Cross-dataset agreement.** Sequences were checked against a sister endpoint's records, themselves validated against INN nomenclature and patent listings: **zero conflicts** across all shared compounds. Two independent derivation paths reaching the same string is evidence against transcription artefact.

**Sequences are recorded for 200 of 259 oligos (77 %) and were never guessed.** The remainder are proprietary, class-level aggregates, or development-code compounds with no published sequence; they carry `TBD`.

## 2.2 Location of chemical modifications

Modification location is captured at three levels of resolution, according to what each source printed: `modification_map` holds **per-residue notation exactly as printed** (e.g. `mG*mC*mG*mA*...`, where `*` marks a phosphorothioate linkage and a leading letter a 2'-modified residue); `gapmer_design` gives wing-gap-wing boundaries (e.g. `5-10-5_MOE`) for 171 compounds; and `sugar_modifications` plus `ps_count` record which chemistries are present and the backbone load (237 and 203 compounds respectively). Where a source printed residue-level notation it is preserved verbatim rather than normalised away, because `sequence_5to3` holds bases only and would otherwise discard the positional detail.

Two encoding hazards were met and are recorded so others avoid them. **Letter case is significant** — it distinguishes modified wings from the DNA gap — so case must be read from the PDF text layer, never judged from rendered pixels. And in one source **chemistry is encoded by font colour and PS/PO by underline**; colour was recoverable from text-layer span attributes, underline was not, so those records retain `TBD` rather than a guess.

## 2.3 Purity data — withheld, not missing

The Challenge requires the dataset to contain **"data on the purity and characterization of each"** oligo. **No numeric purity value is reported for any of the 259 compounds, and none has been inferred, estimated, or carried across from another compound or batch.** An earlier version of this document called that a structural limit of curation. That was imprecise, and the sharper answer matters because it tells a consumer what the gap actually is.

**The purity specification exists and is deliberately not public.** Twelve EMA EPARs (about 3.9 MB of unredacted text) and eleven FDA review packages were downloaded and read in full for this question. Two exhaustive sweeps for any purity, assay or full-length-product term adjacent to a percentage or an NLT/NMT qualifier returned **zero** numeric acceptance criteria. The reason is stated by the regulators themselves: the criteria are Confidential Commercial Information. FDA withholds them under FOIA Exemption 4 with citable page-count stamps — nusinersen NDA 209531, "136 Page(s) has been Withheld in Full as b4 (CCI/TS)"; defibrotide NDA 208114, 125 pages; pegaptanib NDA 21-756, 56 pages and then 46. For imetelstat NDA 217779, 24 of 156 pages are released and the withheld portion is precisely the section titled "Characterization of Drug Substance and Impurities". In the 1998 fomivirsen review the specification pages are physically removed from the scan. This is not an absence of records; it is an active exemption.

**The characterization method is public, and is now recorded.** The same documents name the specification parameters and the analytical methods behind them, and those are captured for 11 compounds in `purity_method` — for example the Waylivra EPAR's active substance specification lists identification by IP-HPLC-TOF-MS and  $T_m$ , most abundant mass by IP-HPLC-UV-MS, sodium counter-ion by ICP-OES, and assay, purity, specified and unspecified impurities, total degradation products and total impurities all by IP-HPLC-UV-MS; the Tegsedi EPAR names full-length product content as the purity analyte. So the requirement is met on characterization method and unmet on numeric purity, and the dataset now says which.

**What a specification value would and would not have told you.** It describes a released lot against an acceptance criterion. It is not a measurement of the material dosed in any particular trial. Even recovered, it could not be attached to a measurement row as the purity of the tested article without a lot linkage that these documents do not provide. That is recorded as an open scientific question rather than resolved unilaterally.

**Positional chemistry, which is the other half of the requirement, advanced substantially.** Per-residue modification maps now cover 44 compounds against 2 at the start of this round, in a documented lossless notation from which the base sequence, every sugar, every 5-methylcytosine and every backbone linkage can be read back; a QC gate rejects any map that disagrees with its own sequence, phosphorothioate count or backbone column. Identity is independently confirmed for a large share of compounds against a patent sequence listing or a regulatory chemical name with a matching molecular formula — a stronger check than a purity percentage alone, and the route by which several chemistry errors in this dataset were caught.

**The gap carries a plan, not a shrug.** `data/recovery_ledger.csv` records 18 open items (10 critical), each with the sources already searched, a next action, an owner and a stopping criterion. For purity that next action — a FOIA request, or sponsor outreach — is recommended for the team's decision; no contact has been initiated.

## 3. Extraction methods

Extraction ran as a multi-agent pipeline over the verified source list, one agent per source, each returning rows conforming to the published schema. Values were transcribed only from text actually retrieved. Tables were parsed from JATS XML where available, from PDF text layers via PyMuPDF otherwise, and two bitmap-only tables were recovered by rendering the page region at 4× and reading it visually.

**Three table-extraction hazards occurred and are documented because each produces plausible, well-formed, wrong data rather than an error:**

- 1. Line-wrapped sequence cells truncate silently.** A naive `[ACGT]`-run regex returns a plausible short sequence when a cell wraps; one run produced 94 corrupted sequences this way. *Guard:* accept a sequence only against an independently declared length.
- 2. Chemistry encoding is invisible to a renderer (§2.2).** *Guard:* resolve from the text layer; leave `TBD` where it cannot be.
- 3. Header/row misalignment returns the wrong analyte.** In three of five tables in one patent the header row omits the leading row-label cell, so addressing a column by header name is off by one — returning **red-cell counts where platelet counts were intended in 35 of 55 rows**, with entirely plausible magnitudes from the correct study. *Guard:* right-alignment of the numeric block, a units-row assertion, and a magnitude check; all 55 values were then re-verified against an independent manual transcription.

The general rule adopted: **validate every value against something outside the parse** — a declared length, a units row, a second host, or a hand transcription. A parser that is merely self-consistent fails silently.

**No-fabrication policy.** Sequences and toxicity values are never invented or recalled from memory. A value is recorded only if it appears in retrieved text, and every row must name its exact locus; a row whose locus cannot be named is dropped rather than kept with a vague citation. TBD is a correct and expected outcome. Compounds without published platelet data are omitted, not padded with assumed zeros.

## 4. Harmonisation and grading

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All categorical fields use controlled vocabularies published in the schema and enforced by the QC gate. Each measurement carries an ordinal **thrombocytopenia\_grade (0–3)**, CTCAE-aligned for clinical rows with explicit in-vitro analogues so bench and trial evidence share one scale. The rubric is built around the endpoint's bimodal presentation — the common dose-dependent decline versus the rare immune-mediated severe form — so that grading does not average them together.

Three grading conventions are applied uniformly: an **incidence row is graded by the severity of the event**, not by how small the incidence is; an **explicit reported zero is grade 0** and its notes record which grade the band would otherwise have carried, so severe-band cells stay findable; and **bleeding is graded on attribution**, requiring haemorrhage attributable to thrombocytopenia for grade 3. Grades are flagged provisional pending subject-matter review.

**Human and animal evidence are separated** by a derived `subject_class` column computed from study type × species and independently re-derived by the QC gate, which fails the build on disagreement. Pooled multi-species findings are assigned to neither side rather than forced.

## 5. Quality control and verification

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`scripts/qc_thrombo.py` gates every ingestion round and exits non-zero on failure: schema conformance against the published enums; referential integrity (zero orphans); primary-key uniqueness; range checks; **provenance completeness** — no row may have an empty or TBD source identifier, reference or locus; case-insensitive sequence validation (case encodes chemistry, so a case-sensitive check would reject correct rows); `subject_class` consistency; and standing warnings for control-arm rows carrying non-zero grades, abstract-only citations, and Phase 2 content coverage.

**Adversarial verification.** Extracted rows are passed to an independent agent instructed to *refute* them: fetch the cited source and confirm it contains the claimed value at the claimed locus, that the grade follows the rubric, and that the row concerns platelets. Verdicts are applied mechanically, so a rejected row cannot survive by oversight. Two blocks totalling 659 rows have completed — **628 confirmed, 31 corrected, 0 rejected** — one of them by recomputing every value from the paper's raw per-replicate workbook, where every mean, SD and n reproduced exactly. **The remaining rows are curated but unverified**, and the dataset must not be described as fully verified until that completes.

Two further independent checks: a **held-out known-answer test** (the inotersen label parsed and its figures recorded before any extraction ran; all six matched) and **cross-dataset sequence agreement** (zero conflicts).

## 6. Reproducibility

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The pipeline is a single command, `scripts/ingest_thrombo.sh`: harvest → merge lanes in a deterministic precedence order → assemble → re-apply committed verification verdicts → QC gate → rebuild the derived view and generated documentation. Determinism was verified by building twice with inputs supplied in opposite orders and confirming byte-identical output; an earlier order-dependence in conflict resolution was found and fixed this way.

The **full curation record is committed** — every agent's returned rows, the verification verdicts, and the source sweep — so the published tables are reproducible from their inputs, not merely re-checkable against their citations. Rebuilding from that record alone reproduces both CSVs byte-for-byte. Verdicts are keyed to a natural key drawn from row content rather than to row IDs, so verification survives the renumbering that re-assembly causes. All tooling is open (Python 3, PyMuPDF, scikit-learn, standard CSV) and no step depends on a proprietary service.

